

Double-Fine-Tuning Multi-Objective Vision-and-Language Transformer for Social Media Popularity Prediction

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ABSTRACT

Social media popularity prediction aims to predict future interaction or attractiveness of new posts. However, in most existing works, there is a notable deficiency in the effective treatment of numerical features. Despite their significant potential to provide ample information, these features are often inadequately processed, leading to insufficiency of information acquirement. In this paper, we introduce a method, named **Double-Fine-Tuning Multi-Objective Vision-and-Language Transformer (DFT-MOVL)**. To supplement the information in vision-and-language pre-training (VLP), we propose compound text, which is concatenated by numerical data and text. Furthermore, during VLP, a transformer is trained using 3 objectives to ensure thorough feature extraction. Finally, for more generalized prediction, we fine-tune 2 models using different training ways and ensemble them. To evaluate the effectiveness of each mechanism adopted in the proposed method, we conduct an array of ablation experiments. Our team achieve the 3rd place in Social Media Prediction (SMP) Challenge 2023.

CCS CONCEPTS

• **Computing methodologies** → *Machine learning*; • **Information systems** → *Multimedia information systems*.

KEYWORDS

Multimedia; Vision-and-Language Transformer; Fine-tuning; Social Media Popularity Prediction

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1 INTRODUCTION

The exponential surge in the volume of multimedia data uploaded to the public Internet, a considerable portion of which is manifested as posts. Through efficient harnessing of these multimodal posts, artificial intelligence can significantly augment user experiences, escalate commercial profits, and enrich our understanding of the world. Thus, for this motivation, Social Media Prediction (SMP) Challenge 2023 proposes the social media popularity prediction (SMPP) task and publishes the social media prediction dataset (SMPD) [23, 25, 26]. The goal is to develop a method to accurately predict the popularity of each post collected from the Internet.

The complexity of multimedia data lies in its intertwined instances of diverse information types, such as image and text, thereby posing the challenge of effective utilization in unison. In recent years, numerous studies have been undertaken and achieved outstanding performance on SMPP, with approaches for feature extraction and pattern learning from multimodal data range from gradient boosted regression trees [2, 6, 10, 13] to deep neural networks [3, 5, 28]. Furthermore, hybrid methods that amalgamate the two especially stand out [7, 9, 21]. Additionally, with the advancement of vision-and-language pre-training (VLP), a surge in VLP-based methodologies achieving significant outcomes has been observed [4, 20, 27].

However, it is still challenging for multimodal representation of the post. It was noted that many previous studies treated numerical features and image-text pairs in isolation, potentially leading to inadequate information extraction during the pre-training phase. Moreover, previous works usually fuse models trained using different algorithms and features, while the neglect of diverse training purposes, i.e., loss function, may weaken the final performance.

In this paper, we propose a method, named **Double-Fine-Tuning Multi-Objective Vision-and-Language Transformer (DFT-MOVL)**. To enhance our VLP model's understanding to posts, we merge numerical data into image-text pairs to increase information. Then, during pre-training, 3 objectives are adopted to train a transformer, achieving collaborative excavation of multimodal information. Subsequently, we fine-tune 2 models using different training ways and ensemble them for better robustness. This innovative method yields

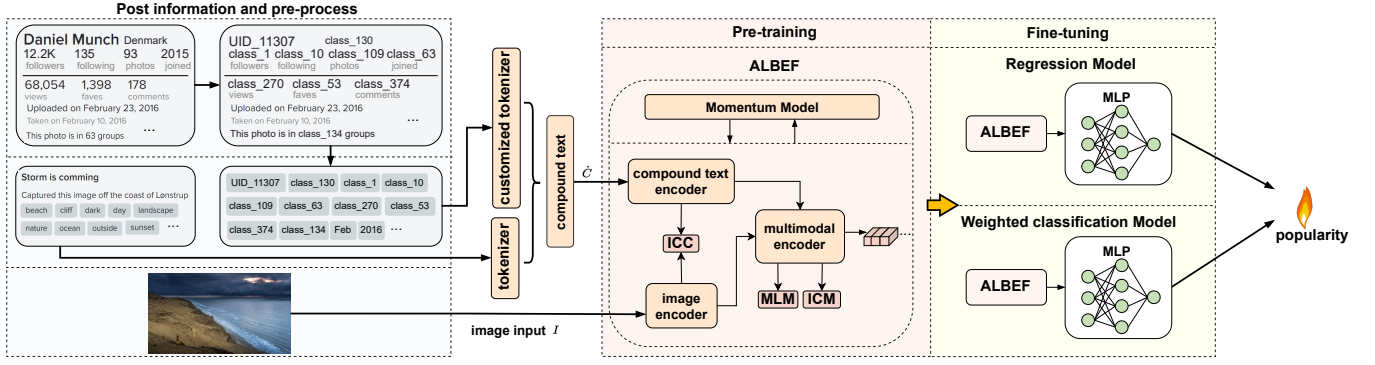


Figure 1: The structure of the DFT-MOVL

significant performance enhancements on SMPD, earning us 3rd place in the SMP Challenge 2023.

Based on the proposed DFT-MOVL, the contributions of the work can be summarized as follows:

- We propose compound text for multimodal pre-training, which consists of numerical features and original text in SMPD. Then we combine compound text with image to form new image-text pairs for further VLP.
- We adopt 3 objectives in our VLP-based transformer for different purposes, including image-compound contrastive learning, masked compound modeling, and image-compound matching.
- We transform SMPP into a weighted classification task. Then we fine-tune 2 models using regression task and weighted classification task, respectively. Finally, we adopt ensemble stacking to fuse results of these 2 models.

The remainder of the paper is organized as follows. Section 2 introduces the related works on SMPP. Section 3 explains our method. Section 4 evaluates the developed method with extensive experiments. Finally, Section 5 concludes this paper and discusses our future work.

2 RELATED WORK

Existing researches on SMPP generally follow a two-step process: feature extraction from multimodal data and subsequent prediction using regression models. The main focus of the research is to employ effective feature extraction methods and regression models to achieve high prediction performance [8–10, 21].

Feature extraction plays a crucial role in capturing the important information that contribute to social media popularity across multimodal data. Numerous methods have been explored for extracting features from image, text, and other modalities, followed by fusion techniques to integrate these modalities. For instance, Wu et al. [24] propose a framework based on multi-scale decomposition and structural reconstruction in a tensor space by joint low-rank constraints. Meghawati et al. [16] propose a multimodal approach to combine social media information using a convolutional model. Chen et al. [3] propose a curriculum learning approach, a wide multimedia-based transformer with graph target detection is employed to learn cross information from different modalities.

Various regression models have been explored to make predictions based on the extracted features. In [2], XGBoost is utilized as the regression model, while the work of [13] uses CatBoost for prediction. A neural network-based regression model is proposed to leverage knowledge from different modalities [28].

Many previous methods for extracting image and text features operate independently, several studies have employed VLP methods, such as ALBEF [14] or CLIP [18], to bridge the gap between image understanding and text comprehension. Tan et al. [20] propose a multimodal data processing method to extract features, the ALBEF pre-trained model is utilized to extract multimodal representations. In [27], a method is proposed to extract deeply exploited features by leveraging several pre-trained models, where the pre-trained CLIP model is employed. In [4], a title-and-tag contrastive vision-and-language transformer model is proposed to discern the distinctions between titles and tags using contrastive learning and 2 pre-trained ViLT models [11]. Further research on VLP methods should be pursued to enhance the performance of SMPP.

3 METHODOLOGY

As shown in Fig. 1, the proposed DFT-MOVL consists of 3 main components: 1) Feature engineering module, in which the combination of numerical data and image-text pairs is prepared for further VLP; 2) VLP-based transformer module, where 3 objectives are adopted for collaborative optimization; 3) 2 models separately fine-tuned using regression task and weighted classification task.

3.1 Feature Engineering for Numerical Data

In SMPP, numerical data of posts represent the most common form of information, offering a wealth of detail. To help the multimodal pre-training model learn patterns and characteristics from posts better, we adopt various feature engineering approaches to extract features from numerical data. The official dataset mainly includes the user profile, which consists of user information, calendar information, and geographical information associated with related posts in the numerical form. Furthermore, in the supplementary information of the post, there is a binary attribute called “Ispublic” which indicates that almost all posts are authenticated with “read” permission. Additionally, according to “pathalias”, i.e., the path alias provided by the user, the user information of most posts is crawled,

such as “totalViews”, “totalFaves”, “totalGroups”, and “followerCount”. Since the popularity prediction is for posts, it is necessary to consider the temporal factors and calculate the “meanViews” and “meanFaves”, rather than just using “totalViews” and “totalFaves”. Combined with the UID (user identifier), user information provides more specific characteristics of the user.

In [23], the popularity metric of social media posts is derived from 2 key parameters: the view count of a photo and the number of days since the photo was posted. Commonly, the view count of a post is influenced by various factors, including the posting time, the number of followers of the user, the number of tags, etc. These numerical information are considered as structured data, distinct from unstructured data like text. We assume that these numerical data can be viewed as a special class of textual data, which will help the multimodal pre-training model with learning the representations of image-text pairs.

To avoid the insufficiency of semantics acquirement, we concatenate these numerical data based on the order of attribute correlations. For instance, the “UID” attribute bears a closer relationship to “UIDPostCount” and “FollowerCount”, while “FirstDate” is closely associated with “FirstDateTaken”. Besides, some numerical data exhibits largely uneven distributions. To deal with it, the data are processed into quantity intervals based on quantiles to ensure an even distribution of data and avoid lots of redundant information. It should be noted that the “UID” is not processed in terms of quantiles, since it is necessary to ensure that each user possesses a unique identifier.

After being processed, the numerical features are added as customized tokens into the vocabulary list of an existing tokenizer. Each numerical feature is treated as a token and concatenated with texts, providing our learnable models with a more comprehensive understanding of the multimodal inputs, thereby enhancing their ability to accurately predict social media post popularity.

3.2 VLP-based Multi-Objective Transformer

To learn better from the image-text data crawled from web sources, which contain a lot of noise, we utilize the ALBEF model to conduct multimodal pre-training. The ALBEF is a VLP method with momentum distillation, aiming to learn multimodal representations. Specifically, momentum distillation is capable to handle the noise existed in web data. ALBEF comprises an image encoder, a text encoder, and a multimodal encoder. In the pre-training objectives, we introduce compound text, which concatenates numerical features with original text. We assume that compound text also contain some kind of semantic information. Three pre-training objectives exist based on ALBEF, where image-compound contrastive learning (ICC) is for the unimodal encoders, while masked compound modeling (MCM) and image-compound matching (ICM) are for the multimodal encoder. We denote image set, text set, numeric set, and the cross-entropy loss as I , T , N , and $H(\cdot)$, respectively. Then we get compound text set $\dot{C} = [T; N]$. Thus, the pre-training objectives with momentum distillation can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{\text{icc}}^{\text{mod}} = & \frac{1-\alpha}{2} \mathbb{E}[H(\mathbf{y}^{\text{i2t}}(I), \mathbf{p}^{\text{i2t}}(\dot{C})) + H(\mathbf{y}^{\text{t2i}}(I), \mathbf{p}^{\text{t2i}}(\dot{C}))] \\ & + \frac{\alpha}{2} \mathbb{E}[\text{KL}(\mathbf{q}^{\text{i2t}}(I) \parallel \mathbf{p}^{\text{i2t}}(\dot{C})) + \text{KL}(\mathbf{q}^{\text{t2i}}(I) \parallel \mathbf{p}^{\text{t2i}}(\dot{C}))], \end{aligned} \quad (1)$$

where $\mathbf{y}$, $\mathbf{p}$, and $\mathbf{q}$ are the ground-truth one-hot similarity, predicted similarity, and predicted similarity from the momentum encoder, respectively.

$$\begin{aligned} \mathcal{L}_{\text{mcm}}^{\text{mod}} = & (1-\alpha) \mathbb{E}[H(\mathbf{y}^{\text{mask}}(I, \dot{C}), \mathbf{p}^{\text{mask}}(I, \dot{C}))] \\ & + \alpha \mathbb{E}[\text{KL}(\mathbf{q}^{\text{mask}}(I, \dot{C}) \parallel \mathbf{p}^{\text{mask}}(I, \dot{C}))], \end{aligned} \quad (2)$$

where $\mathbf{y}$, $\mathbf{p}$, and $\mathbf{q}$ are the one-hot vocabulary distribution, prediction probability for the masked token, and prediction probability for the masked token from the momentum model, respectively.

$$\mathcal{L}_{\text{icm}} = \mathbb{E}[H(\mathbf{y}^{\text{icm}}(I, \dot{C}), \mathbf{p}^{\text{icm}}(I, \dot{C}))], \quad (3)$$

where $\mathbf{y}$ and $\mathbf{p}$ are the 2-dimensional ground-truth one-hot label and prediction, respectively. The full pre-training objective is:

$$\mathcal{L} = \beta \mathcal{L}_{\text{icc}}^{\text{mod}} + \gamma \mathcal{L}_{\text{mcm}}^{\text{mod}} + \omega \mathcal{L}_{\text{icm}}, \quad (4)$$

where α , β , γ , and ω are the weighted of the momentum model, and 3 pre-training objectives, respectively.

To effectively leverage SMPD, We feed the image-compound pairs from SMPD into the ALBEF pre-trained model for incremental pre-training. The pre-training can be expressed as follows:

$$\text{ALBEF}_{\text{Pre-trained}} = \text{ALBEF}_{\text{Base}}(I, \dot{C}), \quad (5)$$

where $\text{ALBEF}_{\text{Base}}$ refers to the pre-trained ALBEF model on a 14M dataset.

3.3 Double Fine-Tuning

We adopt multiple fine-tuning ways based on a Vision-and-Language Transformer to improve the training efficiency and generalizability of our method. As we obtain the pre-trained ALBEF model, we firstly predict the social media popularity prediction using a multilayer perceptron (MLP) on the multimodal encoder’s representation of the [CLS] token. Secondly, we take into account that SMPD is time series correlated, and adding data crawled at the later stage helps but also adds a lot of noise. Therefore, we simplify the complexity of the problem to avoid the influence of noise on the model. Specifically, we transform SMPP into a weighted classification problem by categorizing the existing popularity into 17 classes. Similarly, we predict the class probabilities using an MLP on the representation of the [CLS] token from the multimodal encoder.

$$\text{ALBEF}_{\text{Fine-tuned}} = \text{MLP}[\text{ALBEF}_{\text{Pre-trained}}(I, \dot{C}, p)], \quad (6)$$

where p refers to the ground-truth popularity. By denoting batch size as B , the loss function of the regression, which is Mean Square Error (MSE) can be expressed as follows:

$$\mathcal{L}_r = \frac{1}{B} \sum_{i=1}^B (y_i - \hat{y}_i)^2. \quad (7)$$

The loss function and prediction of the weighted classification are modified as follows:

$$\mathcal{L}_{\text{wc}} = \frac{1}{M} \sum_{k=1}^M e^{-\frac{(t_k - y)^2}{2\sigma^2}} \log p_k, \quad (8)$$

$$\hat{y} = \sum_{k=1}^N t_k p_k, \quad (9)$$

where t_k is the k th classification values of popularity, p_k is the prediction probability of k th class, y is ground-truth popularity, M is the number of classes, and σ is a hyper-parameter.

Then, cross-validation is used to obtain the 768-dimensional multimodal representations of all image-compound pairs based on weighted classification and regression tasks.

Last but not least, to fuse the 2 models, ensemble stacking [22] is adopted in our method, which learns a logistic regression (LR) model for ensemble. At the beginning, we employ 2 MLP-based models to separately generate popularity predictions on the training set to train a LR model. Subsequently, the LR model captures the relationship between the predictions of 2 models and the ground truth. Finally, we employ the LR model to fuse the results of 2 models on the test set, providing a comprehensive popularity prediction.

4 EXPERIMENT

4.1 Dataset, Metrics and Settings

We evaluate our proposed method using the Social Media Prediction Dataset (SMPD), consisting of 486K social media posts from 70K users. The dataset split includes 306K training posts and 180K testing posts. The SMPD is a multi-faced, large-scale, and temporal web data collection. In our experiments, we choose a pre-trained model on a 14M dataset, which contains 3 web datasets (Conceptual Captions [19], SBU Captions [17], and Conceptual 12M dataset [1]), as well as 2 in-domain datasets (COCO [15] and Visual Genome [12]). Then, we utilize the SMPD for incremental pre-training and fine-tuning.

Based on the SMPD, the challenge computes Spearman Ranking Correlation (SRC) and Mean Absolute Error (MAE) for the submitted model as quantitative metrics of performance evaluation.

We incrementally pre-train the model for 10 epochs on 4 NVIDIA GeForce RTX 3090 GPUs with a batch size of 16 and a training duration of 1.5 hours per epoch. We use the AdamW optimizer with a weight decay of 0.02. The learning rate warms up to $1e^{-4}$ in the first 1000 iterations, and declines to $1e^{-5}$ according to a cosine schedule. The parameter for updating the momentum model is set as 0.995, and the queue size for ICC is set as 65536. We linearly ramp up the distillation weight α from 0 to 0.4 within the first epoch. Besides, the hyper-parameters of this works include α , β , γ , ω , and M , set as 0.3, 1, 1, 1, and 17, respectively.

4.2 Result Analysis

The overall experimental results are presented in Table 1. It should be noted that to effectively leverage the dataset, all our experiments in the subsequent sections are performed using cross-validation.

To validate the effectiveness of the proposed method, we implement several common methods for comparison. As shown in Table 1, Method 5-7 exhibits significant improvement compared with Method 1-4, indicating the effectiveness of compound text. Besides, MCM and ICC objectives converge better. Furthermore, by fusing the predicted popularity from the Method 5 and 6, we obtain the final method, DFT-MOVL, with an SRC of 0.753 and an MAE of 1.170, indicating the effectiveness of ensemble of double fine-tuned models.

Besides, by using only the basic image-text dataset for incremental pre-training and fine-tuning (Method 1), the results are the

worst. After incorporating the 3 levels of categories into the text information, Method 2 increases the SRC by 0.053 and decreases the MAE by 0.324. Meanwhile, image-text contrastive learning objective in VLP, demonstrates improved convergence, proving the effectiveness of the category set in capturing view-invariant semantic information. Then, we concatenate multimodal representations with word embedding and normalization preprocessing numerical features as inputs and feed them into MLP layers together for regression (Method 3), which yields worse SRC and MAE. On the other hand, we experiment with compound text that only combines text with UIDs (Method 4), indicating the limitation of UIDs.

Table 1: The ablation study presenting SRC and MAE, where all methods are based on ALBEF using different features. Basic methods used for each result: 1) ALBEF_O: ALBEF method with Official image-text dataset; 2) ALBEF_IC: ALBEF method with image-compound dataset. Additional features used for each result: 1) CF: Categorical features; 2) FE: Features engineered from UIDs and numerical data; 3) U: UIDs. Task type: 1) R: Conduct training in the principle of regression task; 2) WC: Transform the training into a weighted classification task. Note that, DFT-MOVL is our proposed method, combining the results of Method 5 and 6.

No.	Method	Task	Offline	Online
1	ALBEF_O	R	0.684/1.579	0.582/1.722
2	ALBEF_O+CF	R	0.733/1.211	0.635/1.398
3	ALBEF_O+CF+FE	R	0.718/1.384	0.623/1.549
4	ALBEF_O+CF+U	R	0.749/1.173	0.649/1.342
5	ALBEF_IC+CF	R	0.847/0.918	0.747/1.175
6	ALBEF_IC+CF	WC	0.832/0.934	0.743/1.179
7	DFT-MOVL	R+WC	0.859/0.906	0.753/1.170

5 CONCLUSION AND FUTURE WORK

In this paper, we propose a method named DFT-MOVL for social media popularity prediction. Multiple mechanisms are applied in the proposed method. Firstly, we merge numerical data into image-text pairs of SMPD to help the VLP model learn multimodal information better. Secondly, 3 objectives are adopted for VLP-based transformer to generate more comprehensive representations. Thirdly, for better robustness, we develop and ensemble 2 fine-tuned models, including an MLP-based regression model and an MLP-based weighted classification model. Last but not least, we conduct exhaustive experiments on the official dataset and win the 3rd place in the SMP Challenge 2023. In the future, we will integrate the proposed method with the Mixture-of-Experts (MoE) layer, which is expected to significantly increase the training efficiency.

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